E-news Express

Project Business Statistics: Enews_Express

Oct 03, 2024

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Executive Summary

E-news Express, an online news portal, aims to expand its business by acquiring new subscribers. The executives of E-news Express are of the opinion that there has been a decline in Monthly subscribers over the past year. The company attributes this drop to the design and content of its current landing page, Which may not be effectively engaging users. The Objective of this project is to determine whether the new landing page is more effective than the old one in terms of user behavior and conversion rate . Using an A/B Testing, 100 users were randomly divided into two groups: Control Group, served with the old landing page and Treatment Group, served with the new landing page.

The analysis of E-news Express´s data revels that the users spend significantly more time on the new landing page compared to the old one, then the Conversion rate for the new landing page is significantly higher, the Conversion status is independent of the user´s preferred language and the the time spent on the new page amoung users with different language preference, suggesting potencial areas for Customization.

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Executive Summary

Conclusions and Recommendations:

The data analysis reveals several key insights about users' preferences and behaviour. Analysis using a two-sample t-test showed that we have enough statistical evidence to say that the mean of the average time spend on the new landing page is greater than the average time spend on the existing one. This suggest that the new landing page design is more engaging, providing a better user experience. The proportion z-test confirmed that the new landing page is more effective in converting users into subscribers. Then, a chi-square test indicate that there is no significant association between the user´s preferred language and their conversion rate, both are independent. Then, a one-way ANOVA test revealed that there are not significant differences in the average time spent on the new landing page across different language groups. Then, the new landing page has been proven to be more engaging and effective at converting. There is an opportunity to refine the page further by considering specific preferences of users based on their language to optimize time spent on the page and maximize engagement for all groups.

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Business Problem Overview and Solution Approach

Business Problem Overview

E-news Express, an online news portal, is facing a decline in subscribers. The management suspects that the current landing page may not be engaging enough, leading to reduced user interactions and conversions. To address this issue, the company has redesigned the landing website with new content and layout to improve user engagement.

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Business Problem Overview and Solution Approach

Solution Approach and Methodology

The solution approach for evaluating the effectiveness of the new landing page design involve a structured analysis using data and statistical testing. To tackle the problem, I analyzed the provided dataset (abtest.cvs), focusing on user interactions with both the old and new landing pages, to analyze whether a new feature attracts users. I used descriptive statistics for key variables (time_spent_on_the_page, converted) to understand the general trends in the data. I used visual analysis, bar charts, boxplots, and histograms to visualize the distribution of time_spent_on_the_page across different groups of languages, conversion rates for the old vs. new landing page and relationship between conversion status and language preference. Additionally, I define the null and alternative hypotheses for each research question and selected appropriate statistical test, t-test, z-test, chi-square test and one-way ANOVA test based on the type of data and hypothesis and calculate the p-values for each hypothesis and determine whether to reject or fail the null hypothesis at a 5% significance level. Based on these insights, I proposed conclusions and recommendations.

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EDA Results

Key Results from **Univariate Analysis**:

Time Spent on the Page: The distribution of the time spent on the page is fairly symmetrical, with most users spending between 4 to 8 minutes on the page. The bloxplot reveals that the median time spent is around 6 minutes, and there are few outliers on the higher end, indicating some users spent significantly more time on the page.

Group Distribution: The group variable split between the control and treatment groups, with each having 50 users, this balanced distribution ensures a fair comparison between the old and new landing page.

Landing Page Distribution: landing_page variable shows an even split, with 50 users assigned to the old landing page and 50 users to the new landing page, it confirms that the A/B testing was set up correctly with equal representation.

Link to Appendix slide on data background check

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EDA Results

Key Results from **Univariate Analysis**:

Conversion Status: Out of the 100 users, 54 converted and 46 did not converted. The count plot shows a slightly higher count of user who converted, suggesting that the overall conversion rate is 54%.

Preferred Language Distribution: is relative balanced among the three languages: Spanish and French with 34 users each and English with 32 users. This balance ensures language-based analysis is not skewed toward any particular group.

Link to Appendix slide on data background check

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EDA Results

Key Results from **Bivariate Analysis**:

Conversion Status vs. Time Spent on the Page: Users who converted (subscribed) spent significantly more time on the landing page compared to those did not convert. The median time spent for converted users is around 7 minutes, while for non-converted users it is around 5 minutes.

Landing Page vs. Time Spent on the Page: Users on the new landing page tend to spend more time compared to those on the old landing page. The median time spent is higher for the new page, indicating that the new design is more engaging. The Distribution is also wider for the new landing page, suggesting that users have varied engagement levels dependent on the content.

Language Preferences vs. Time Spent on the Page: Users with different language preferences show varying time spent on the page. English speakers tend to spend more time on the page compared to Spanish and French speakers. This variation suggest that language preference influences user engagement and may require adjustments based on language.

Link to Appendix slide on data background check

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Hypotheses Tested and Results

1) Hypothesis: Time Spent on the New vs. Old landing page

Null Hypothesis (Ho): There is no significance difference in the average time spend on the new landing page compared to the existing landing page : $\mu new =< \mu old$

Alternative Hypothesis (Ha): Users spend significantly more time on the new landing page compared to the existing landing page: $\mu new > \mu old$

Statistical test: Two-sample independent t-test

Test Result: The calculated p-value from the test is 0.0001 , lower than 0.05

Conclusion: There is sufficient statistical evidence to conclude that the average time spent on the new landing page is higher than the average time spent on the old landing page. This suggest that the new design is more engaging and effective in holding the user´s attention.

Link to Appendix slide on details of the test performed

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Hypotheses Tested and Results

1) Hypothesis: Time Spent on the New vs. Old landing page

Visual Analysis Performed: Boxplot shows that users spend more time on average on the new landing page compared to the old one. The median time spent is visibly higher for the new page, and the interquartile range is wider, indicating greater engagement with the new design.

Final Inference: The results confirm that the new landing page is significantly more engaging, as indicated by the longer average time spent on it. The Statistical test supports the visual analysis, making it reasonable to recommend adopting the new design to replace the old landing page for better user retention and engagement.

Link to Appendix slide on details of the test performed

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Hypotheses Tested and Results

2) Hypothesis: Conversion Rate for the New vs. Old Landing Page

Null Hypothesis (Ho): The conversion rate for the new landing page is lower than or equal to the conversion rate for the old landing page: Pnew =< Pold

Alternative Hypothesis (Ha): The conversion rate for the new landing page is greater than the conversion rate for the old landing page : Pnew > Pold

Statistical test: One-tailed proportion z-test

Test Result: The calculated p-value from the test is 0.008 , lower than 0.05

Conclusion: There is sufficient statistical evidence to conclude that the conversion rate for the new landing page is significantly higher than the conversion rate for the old landing page. This indicate that the new design is more effective at converting users into subscribers.

Link to Appendix slide on details of the test performed

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Hypotheses Tested and Results

2) Hypothesis: Conversion Rate for the New vs. Old Landing Page

Visual Analysis Performed: Bar Plot: Conversion Rate by Group (Control vs. Treatment): The Bar Plot shows that the proportion of users who converted is higher in the treatment group compared to the control group. This visual evidence is supported by the hypothesis test result.

Final Inference: The statistical test results confirm that the new landing page has a significantly higher conversion rate. This suggest that adopting the new landing page could help address the decline in subscriptions and improve overall user conversions.

Link to Appendix slide on details of the test performed

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Hypotheses Tested and Results

3) Hypothesis: Conversion Status vs. Preferred Language

Null Hypothesis (Ho): The conversion status is independent of the user´s preferred language, there is not relationship between language preference and whether a user converts.

Alternative Hypothesis (Ha): The conversion status is dependent of the user´s preferred language, there is a relationship between language preference and whether a user converts.

Statistical test: Chi-square test of independence

Test Result: The calculated p-value from the test is 0.212 , higher than 0.05

Conclusion: There is no sufficient statistical evidence to conclude that the conversion status depends on the preferred language. This implies that the new landing page is equally effective for users regardless of their preferred language.

Link to Appendix slide on details of the test performed

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Hypotheses Tested and Results

3) Hypothesis: Conversion Status vs. Preferred Language

Visual Analysis Performed: Bar Plot: Conversion Status by Preferred Language: The Bar Plot shows the conversion rate is similar across different language groups. The lack of major differences visually supports the conclusion that language does not significantly influence the likelihood of conversion.

Final Inference: The statistical test results confirm that the conversion status is independent of the preferred language. This indicates that the new landing page design is effective across different language groups.

Link to Appendix slide on details of the test performed

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Hypotheses Tested and Results

4) Hypothesis: Time Spent on the New Page for Different Language Users

Null Hypothesis (Ho): The mean time spent on the new landing page is the same for different language users: μEnglish = μNon-English

Alternative Hypothesis (Ha): The mean time spent on the new landing page differs between English and Non-English users: μEnglish ≠ μNon-English

Statistical test: One-way ANOVA test

Test Result: The calculated p-value from the test is 0.43 , higher than 0.05

Conclusion: There is no sufficient statistical evidence to conclude that the mean time spent on the new landing page differs between different languages groups. This suggests that the new landing page is equally engaging for English, French and Spanish.

Link to Appendix slide on details of the test performed

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Hypotheses Tested and Results

4) Hypothesis: Time Spent on the New Page for Different Language Users

Visual Analysis Performed: BoxPlot: Time Spent on the New Page by Language. The bloxplot shows that the distribution of time spent is relatively similar for English, French and Spanish users.

Final Inference: The statistical test results confirm that there is no significant difference in the mean time spent on the new landing page across language groups. This suggests that the new landing page performs well across different languages. This analysis supports the effectiveness of the new landing page for a diverse user base.

Link to Appendix slide on details of the test performed

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APPENDIX

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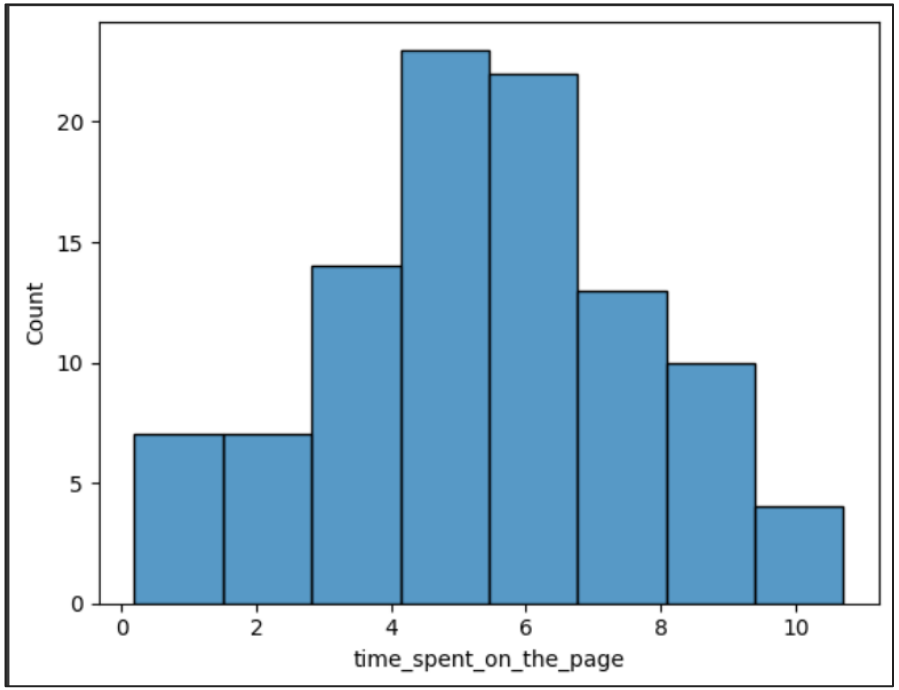
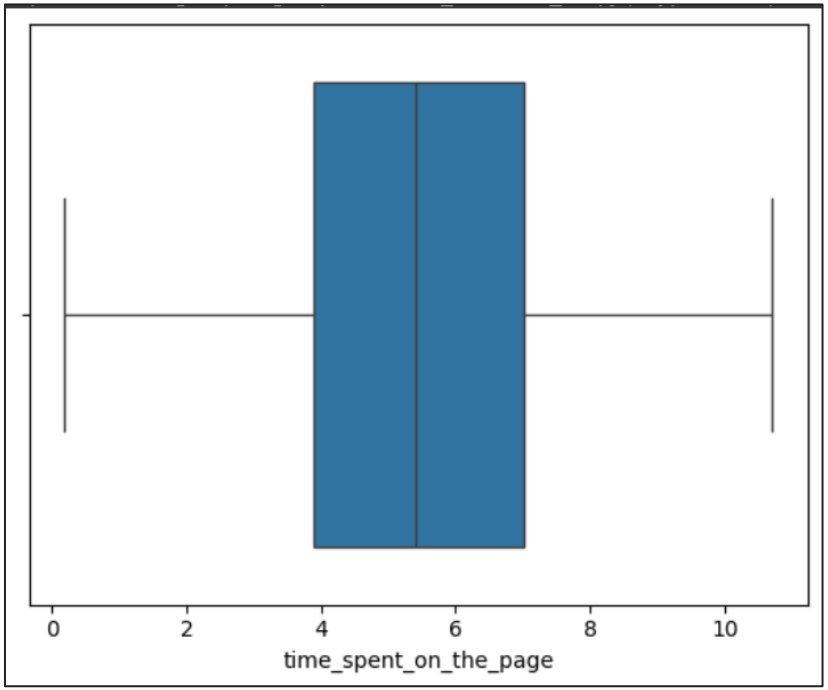
Data Background and Contents

Univariate Analysis: Time_Spent_on_the_page





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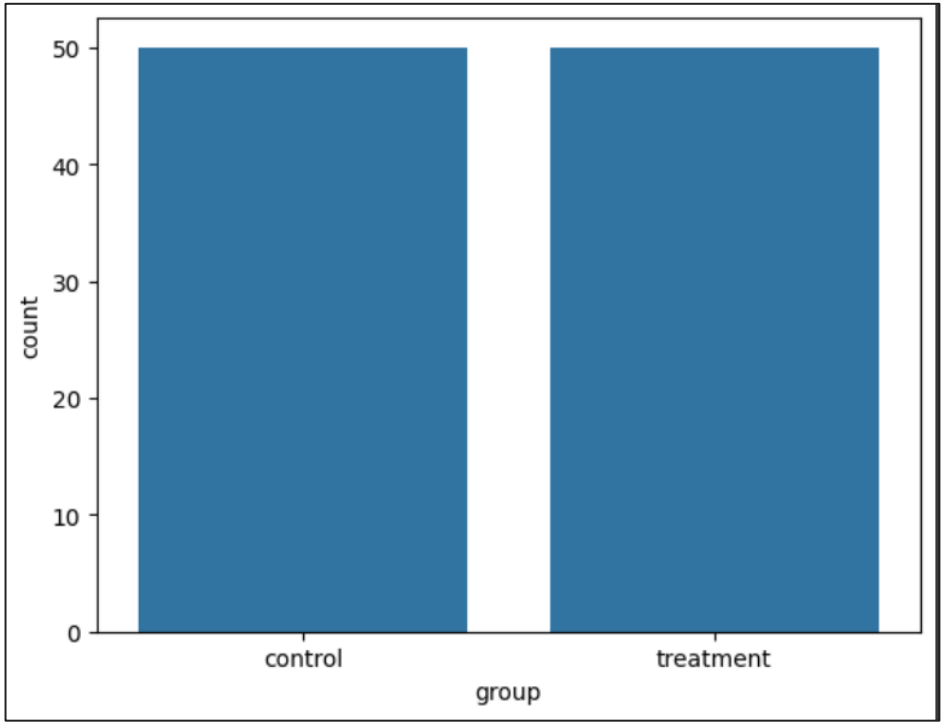
Data Background and Contents

Univariate Analysis: Control group vs treatment group



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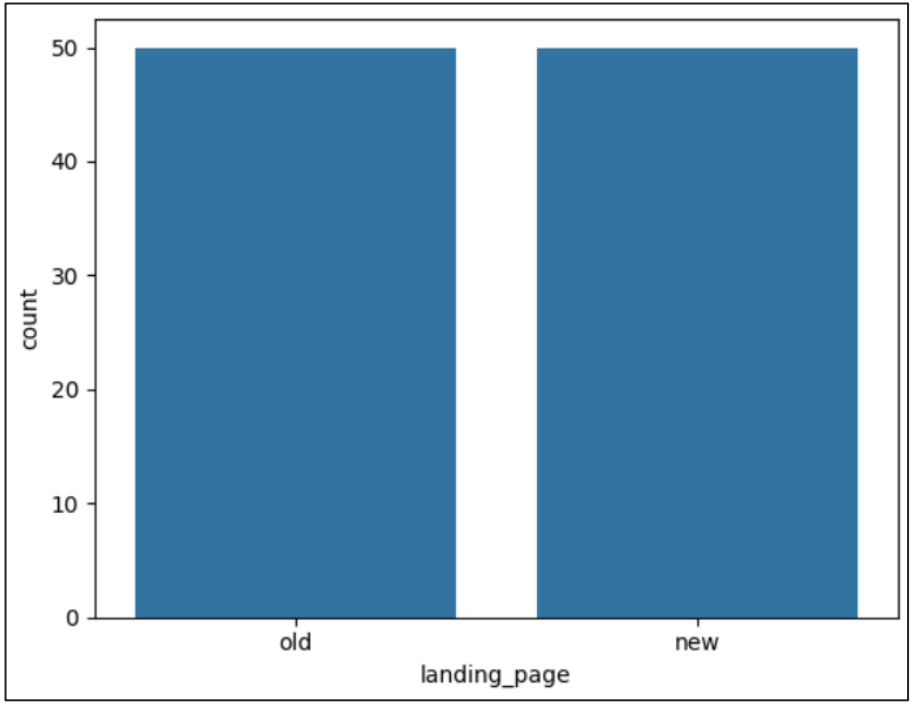
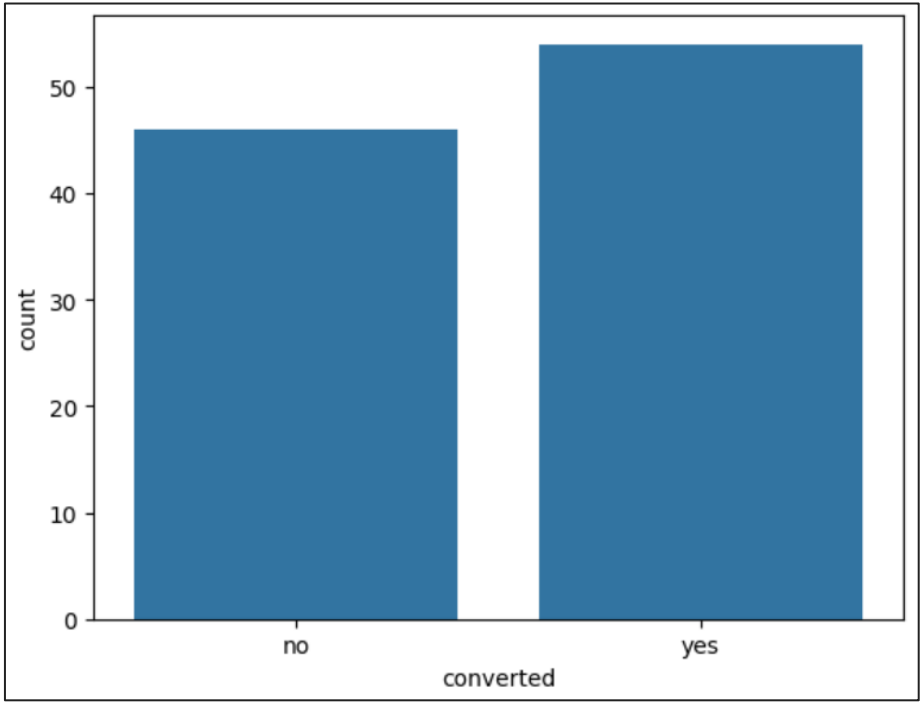
Data Background and Contents

Univariate Analysis: landing_page



Univariate Analysis: Converted



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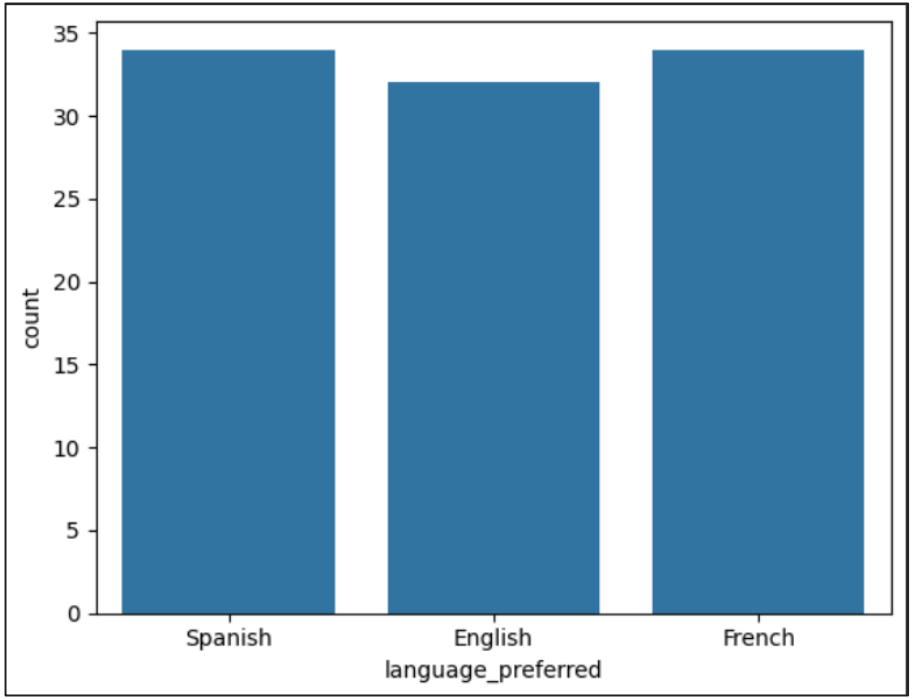
Data Background and Contents

Univariate Analysis: Language preferred



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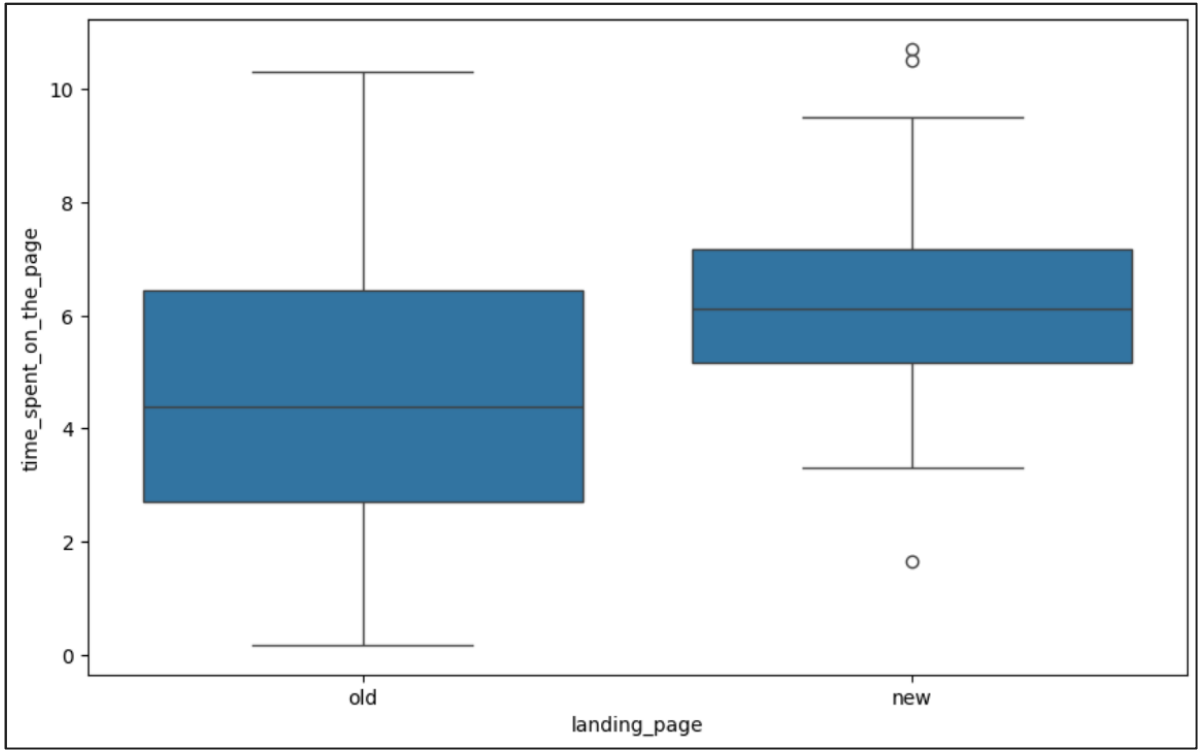
Data Background and Contents

Bivariate Analysis: Landing page vs. Time Spent on the page



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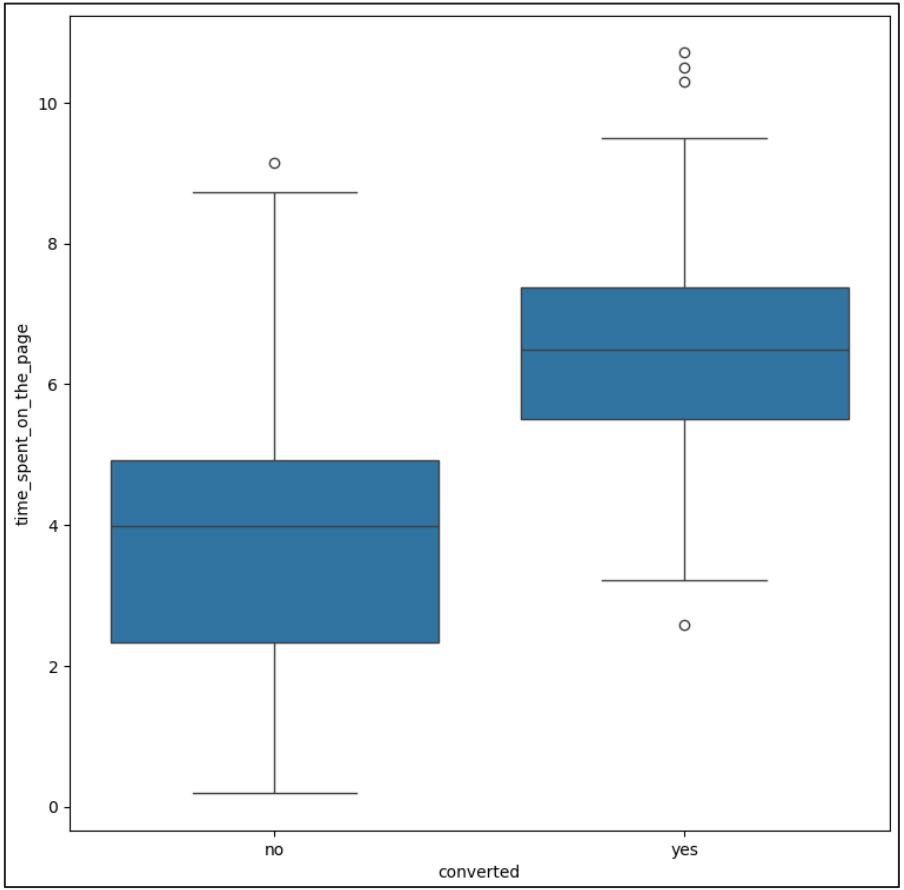
Data Background and Contents

Bivariate Analysis:

Conversion status vs. Time Spent on the page



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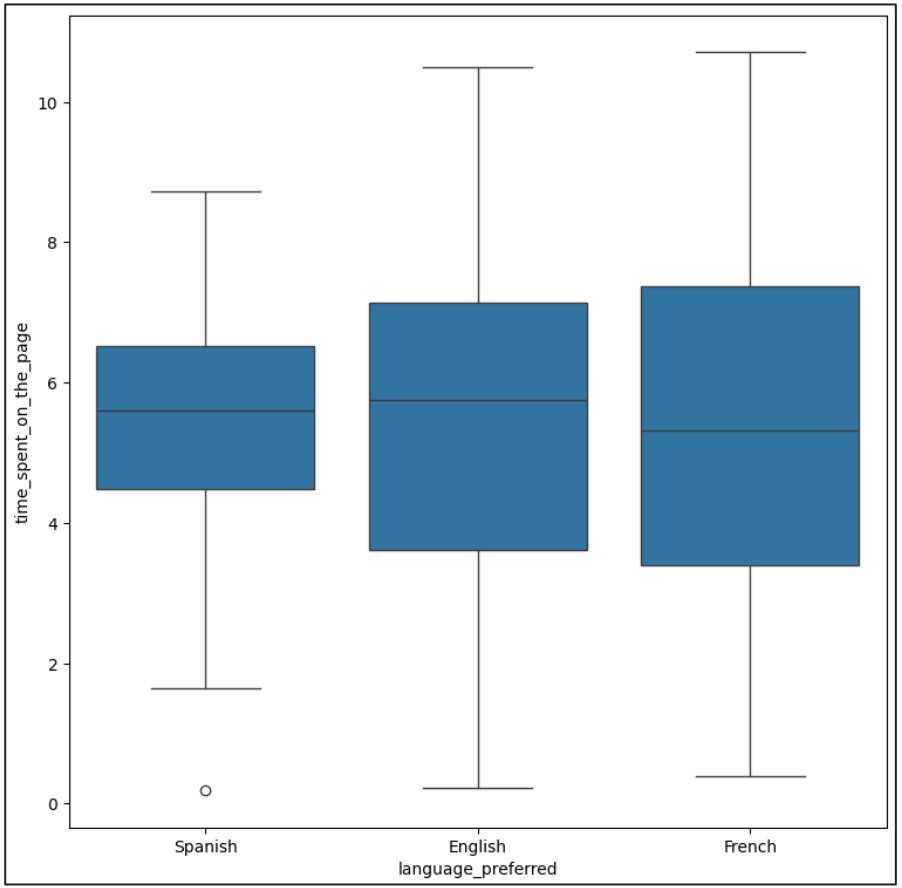
Data Background and Contents

Bivariate Analysis:

Language preferred vs. Time Spent on the page



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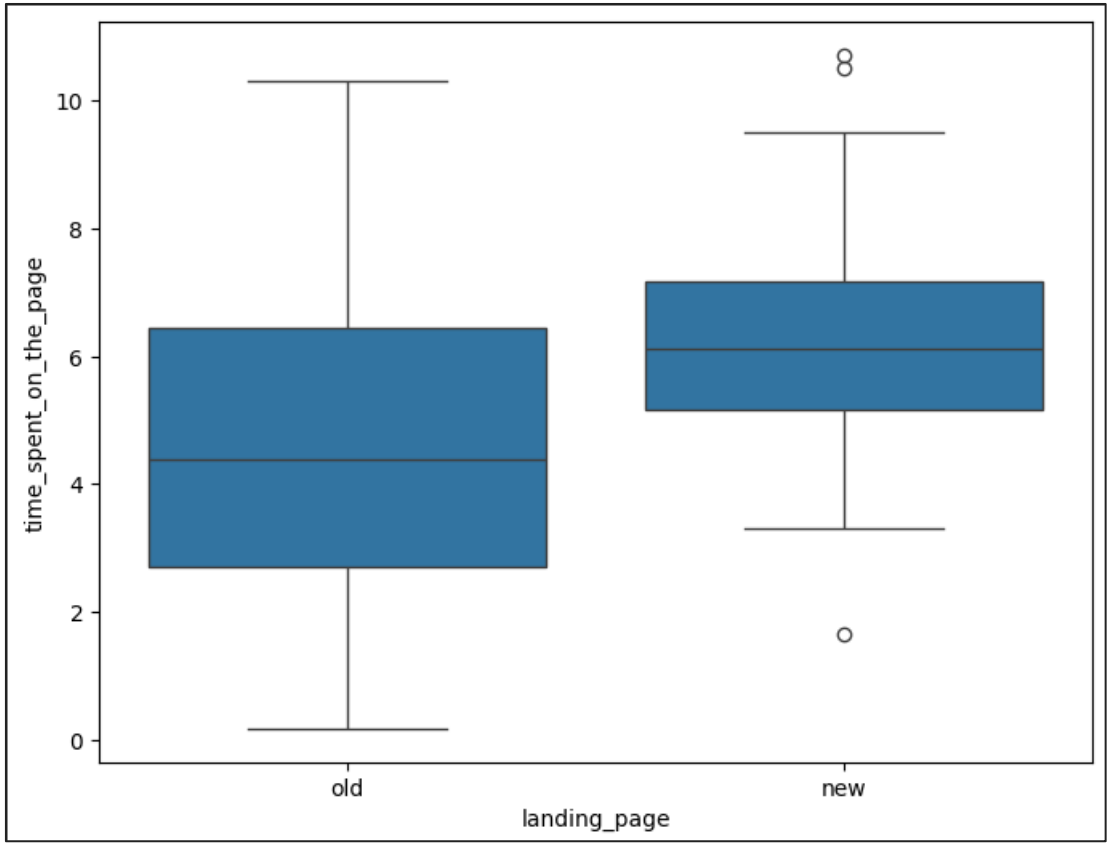
Hypothesis Testing Details

1) **Do the users spend more time on the new landing page than the existing landing page?**

Statistical test: Two-sample independent t-test

Test Result: The calculated p-value from the test is 0.0001 , lower than 0.05

Conclusion: There is sufficient statistical evidence to conclude that the average time spent on the new landing page is higher than the average time spent on the old landing page. This suggest that the new design is more engaging and effective in holding the user´s attention.



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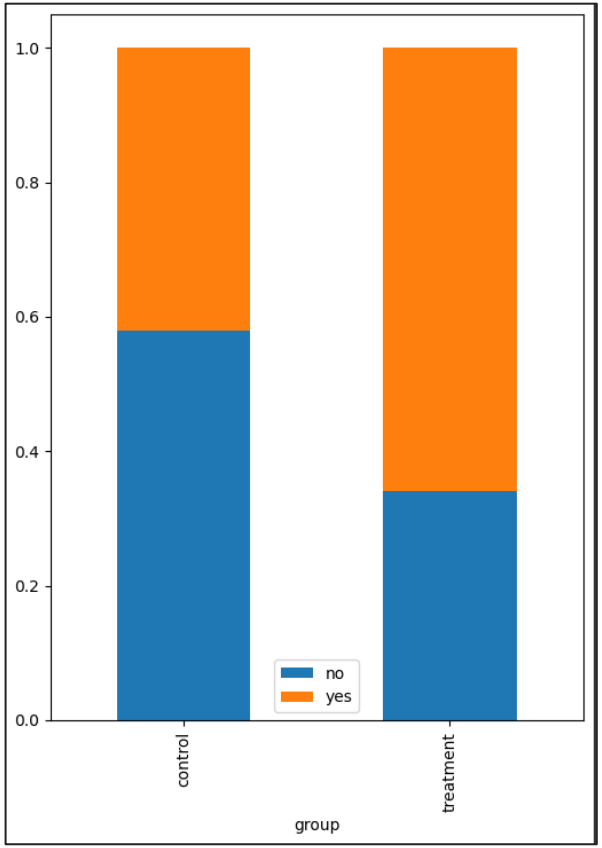
Hypothesis Testing Details

2) Is the conversion rate for the new page greater than the conversion rate for the old page?

Statistical test: One-tailed proportion z-test

Test Result: The calculated p-value from the test is 0.008 , lower than 0.05

Conclusion: There is sufficient statistical evidence to conclude that the conversion rate for the new landing page is significantly higher than the conversion rate for the old landing page. This indicate that the new design is more effective at converting users into subscribers.



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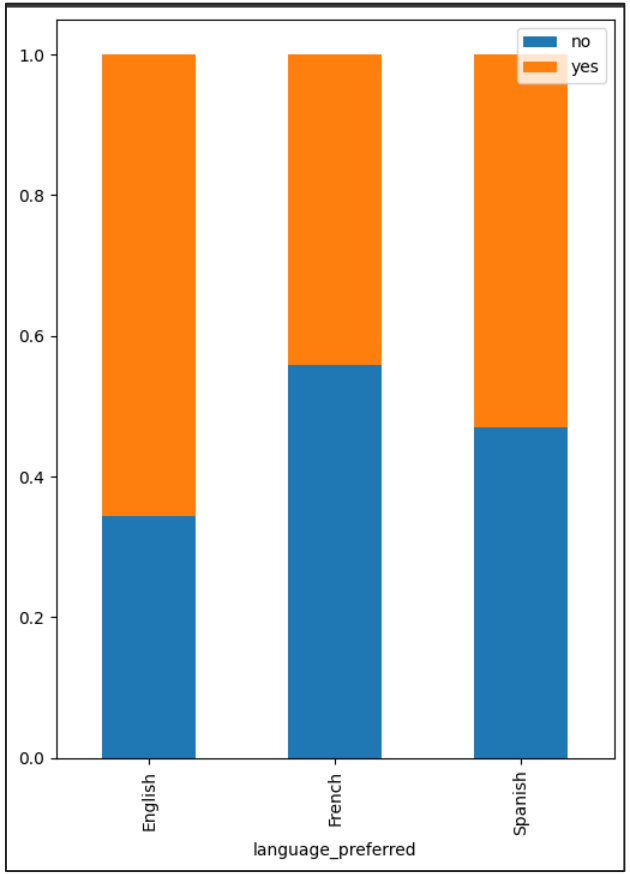
Hypothesis Testing Details

3) Does the converted status depend on the preferred language?

Statistical test: Chi-square test of independence

Test Result: The calculated p-value from the test is 0.212 , higher than 0.05

Conclusion: There is no sufficient statistical evidence to conclude that the conversion status depends on the preferred language. This implies that the new landing page is equally effective for users regardless of their preferred language.



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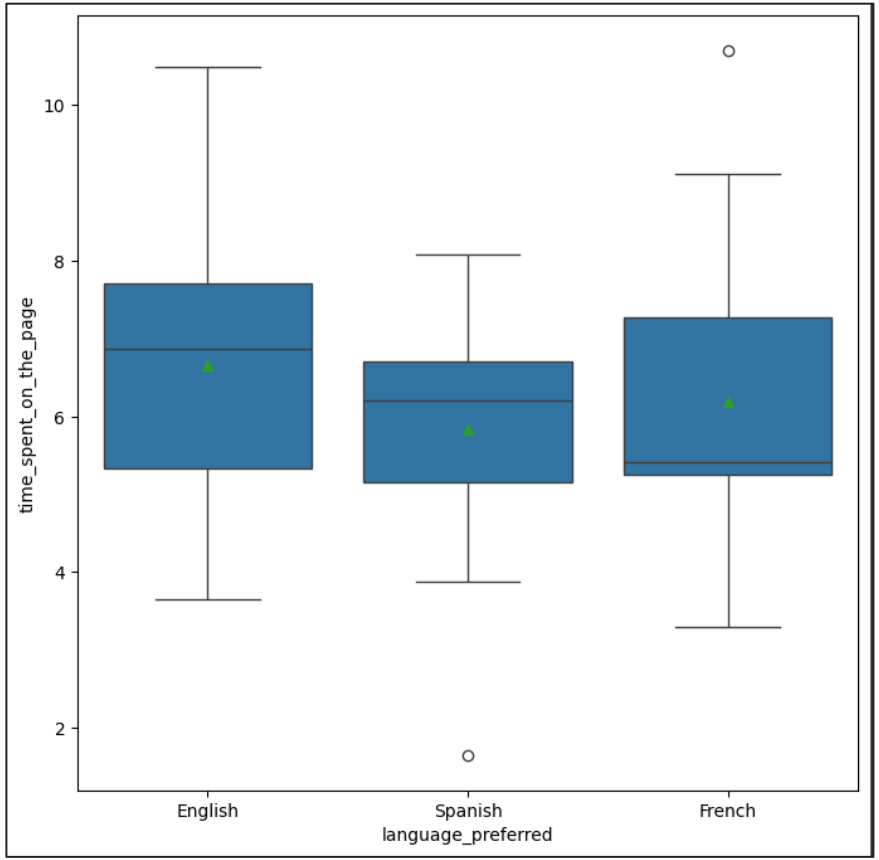
Hypothesis Testing Details

4) Is the time spent on the new page same for the different language users?

Statistical test: One-way ANOVA test

Test Result: The calculated p-value from the test is 0.43 , higher than 0.05

Conclusion: There is no sufficient statistical evidence to conclude that the mean time spent on the new landing page differs between different languages groups. This suggests that the new landing page is equally engaging for English, French and Spanish.



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